Time Series Modeling for Monthly Average Retail Prices of the Samba Rice in Colombo, Sri Lanka.

Project Report

MFM 3151 - Case Study II

Group No. 13

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Supervisor

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January 31, 2025

DECLARATION

I'm E.M.C.G Ekanayake, declare that the presented project report titled, " Time series modeling for monthly average retail prices of the Samba rice in Colombo, Sri Lanka." is uniquely prepared by me based on the group project carried out under the supervision of Ms. K.G.P.Hansani, Department of Mathematics, Faculty of Science, University of Ruhuna, as a partial fulfillment of the requirements of the level III, Case Study II course unit, MFM 3151 of the Bachelor of Science Honours Degree in Financial Mathematics and industrial Statistics in Department of Mathematics, Faculty of Science, University of Ruhuna, Sri Lanka.

It has not been submitted to any other institution or study program by me for any other purpose.

Signature: . . . .
Date: . . . .

SUPERVISOR'S RECOMMENDATION

I certify that E.M.C.G Ekanayake carried out this study under my supervision.

Signature:. . . . .
Date:. . . . .
Department of Mathematics
Faculty of Science
University of Ruhuna.

Abstract

This study deployed time series models to forecast the monthly average retail price of Samba rice in Colombo, Sri Lanka. Given rice's importance to Sri Lanka's economy, reliable price predictions are critical for market regulation and planning. Data preparation includes addressing missing values and converting the data to ensure stationarity, which was confirmed using the Augmented Dickey-Fuller (ADF) test. ARIMA models were used both manually and automatically, with performance evaluated using AIC, BIC, RMSE, and the Ljung-Box test. The Auto ARIMA model (ARIMA (0,1,1)) outperformed the manual model (ARIMA (2,1,1)), with lower AIC (688.19 vs 688.63) and BIC (694.16 vs 700.57) values, indicating improved predictive accuracy. The Ljung-Box test proved that both models did not have residual autocorrelation. The 12-month prediction using the Auto ARIMA model is useful for policymakers and market players, allowing them to make more informed decisions and maintain market stability. This study highlights the efficacy of automated ARIMA approaches for commodity price forecasting, which may be extended to anticipate rice prices following rapid economic shifts.

Keywords: AIC, ARIMA, BIC, Forecasting, Rice Price

Acknowledgment

First of all, my sincere gratitude goes to our supervisor Ms. K.G.P.Hansani for her unending guidance and support for us to succeed in this research. Moreover, I extend my thanks to all the instructors who helped me to make this research a success. Also, I express my sincere appreciation to my team members for sharing their knowledge with me.

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1 Introduction

1.1 Background of the Study

Sri Lanka is an agricultural country with a long history of farming, and rice has always formed the centerpiece of its agriculture. Being the staple foodstuff of the people of Sri Lanka, rice assumes prime importance in the daily lives of Sri Lankans. It forms part of the three square meals of the day, which is one of the essential items in the country. Of the various types, samba rice is the more consumed one; it is also a vital part of the diet amongst Sri Lankans. The importance of rice is more than that of a kind of food. It is directly related to the agronomic economy of the nation. A significant number of people depend on rice cultivation, and it plays a crucial role in the economic stability of the country as a whole. Therefore, it is not only the interest of consumers that requires learning about retail prices but also for the whole economy. The prices of rice in Sri Lanka have recently been very volatile. Retail prices have changed quite frequently due to a number of factors including issues over production, import policies, and external market conditions. These frequent fluctuations have caused concern among both consumers and policymakers. Forecasting retail prices would give farmers, traders, and government authorities much-needed insights for better decision-making and thereby stabilizing the market. Thus, this study kept a close eye on the forecast of retail prices of most consumer samba rice in Sri Lanka. By understanding the trend, we could predict future movements that will help in planning and managing the rice supply chain more efficiently. Given the prime place of rice in Sri Lankan agriculture, this study will discuss in detail the pattern of retail prices and offer certain remedies to dampen the shocks of price variability, considering the issues relating to price stability at present.

1.2 Problem statement

Significant volatilities in retail prices of samba rice in Sri Lanka have posed serious problems to the farmers, consumers, and policy planners. Frequent fluctuations arise on various factors such as production issues, import policies, and external market conditions that complicate the decision-making process. The present study was designed to apply time series analysis in the forecast of retail prices of white rice. The results will provide useful insights that can help in the stabilization of the market and ensure food security, thus supporting strategic planning for the stakeholders in the rice supply chain.

1.3 Significance of the Study

Accurate forecasts of retail rice prices are essential for the stability of the Sri Lankan market and help farmers make well-informed decisions about when to sow and sell their paddy. Forecasting models provide valuable predictions by using data on trends, seasonality, cyclic patterns, and irregular variations. This enables policymakers to create effective agricultural policies and support farmers and rural development. Additionally, to ensure a stable rice supply for consumers, officials can proactively address food security issues by projecting future pricing patterns. Accurate rice price estimates are also crucial for Sri Lanka.

2 Literature Review

Many studies have used time series models to predict prices and production in agriculture. These models are helpful for farmers, traders, and policymakers to make better decisions.For example, Mitra and Paul (2021)[1] applied the ARFIMA model to forecast the daily wholesale rice prices in India. They compared ARFIMA with ARIMA and found that ARFIMA performed better because it captured long-term patterns in the data. This study highlighted the importance of choosing the right model for accurate forecasts.

Jadhav et al. (2017)[2] used ARIMA to predict prices of paddy, ragi, and maize in Karnataka, India. They showed that ARIMA could effectively forecast cereal prices by identifying patterns in historical data. The model helped farmers make better production and marketing decisions, proving its usefulness in planning for price fluctuations. Balilla et al. (2023)[3] focused on the retail prices of eggs, rice, and onions in the Philippines. They used both ARIMA and SARIMA models and found them effective for predicting short-term and seasonal trends. Their study demonstrated how these models could handle seasonal variations and provide reliable forecasts.De Silva and Yamao (2009)[4] used time series forecast models to show that cultivated and harvested land had increasing trend. Such a trend resulted to upward movement in retail rice prices. The studies conducted by these authors corroborate the use of time series models, like ARIMA, in price and fluctuation of production over time.

Based on the above mentioned studies, the present research would like to identify past and current trend of the Retail Price of Samba rice in Sri Lanka. Its purpose is to construct an an appropriate time series model to predict future retail prices. They include market structure, national rice exports, government policies, and food prices etc This research aims at applying these approaches to the Samba rice consumption market and help policy makers control prices in the Sri Lankan market.

3 Aims and Objectives

In this work our aims are to

- Analyze monthly average retail price of samba rice in Colombo by using time series

- Formulate a model to forecast monthly average retail price of samba rice by using the historical data

4 Methodology

4.1 Research Approach

With the concern about the monthly average retail price of samba rice in colombo, outsiders will be able to get an idea about future retail price of samba rice. When pricing done with the economic fluctuation this will be a plus point for both manufacturers and consumers. So, our approach is to determine that model using prior collected data.

4.2 Data Set

For the above mentioned requirement here, we used a data set from "Humanitarian Data Exchange (HDX)" web site which linked below.

Link : https://data.humdata.org/dataset/wfp-food-prices-for-sri-lanka?

4.3 Methodology

4.3.1 Data preparation

We consider only samba rice and only data from Colombo city. There are 192 observations. The dataset was divided into two parts: 80% for training and 20% for testing. The training dataset contains 153 observations, while the testing dataset includes the remaining 39 observations. Our time series model was built using the training data, enabling us to forecast future values. These forecasted values can then be compared with the actual values to evaluate the model's accuracy

Link : https://drive.google.com/file/d/1YQM_0_52XmBpRHZ1dYhxVu71xDEeYKDL/view?usp=drive_link

Variables	Unit	Description	Type
Date	MM/DD/YYYY	The specific date of the price record	Date
Price	Sri lankan Rupees(LKR)	Retail price of samba rice in Sri Lankan rupees	Numerical

Table 1: Variable description

The above mentioned data set will analyze mainly by using **R statistical software**.

4.3.2 The Conceptual Model

The conceptual model presented below aids in understanding the research methodology.

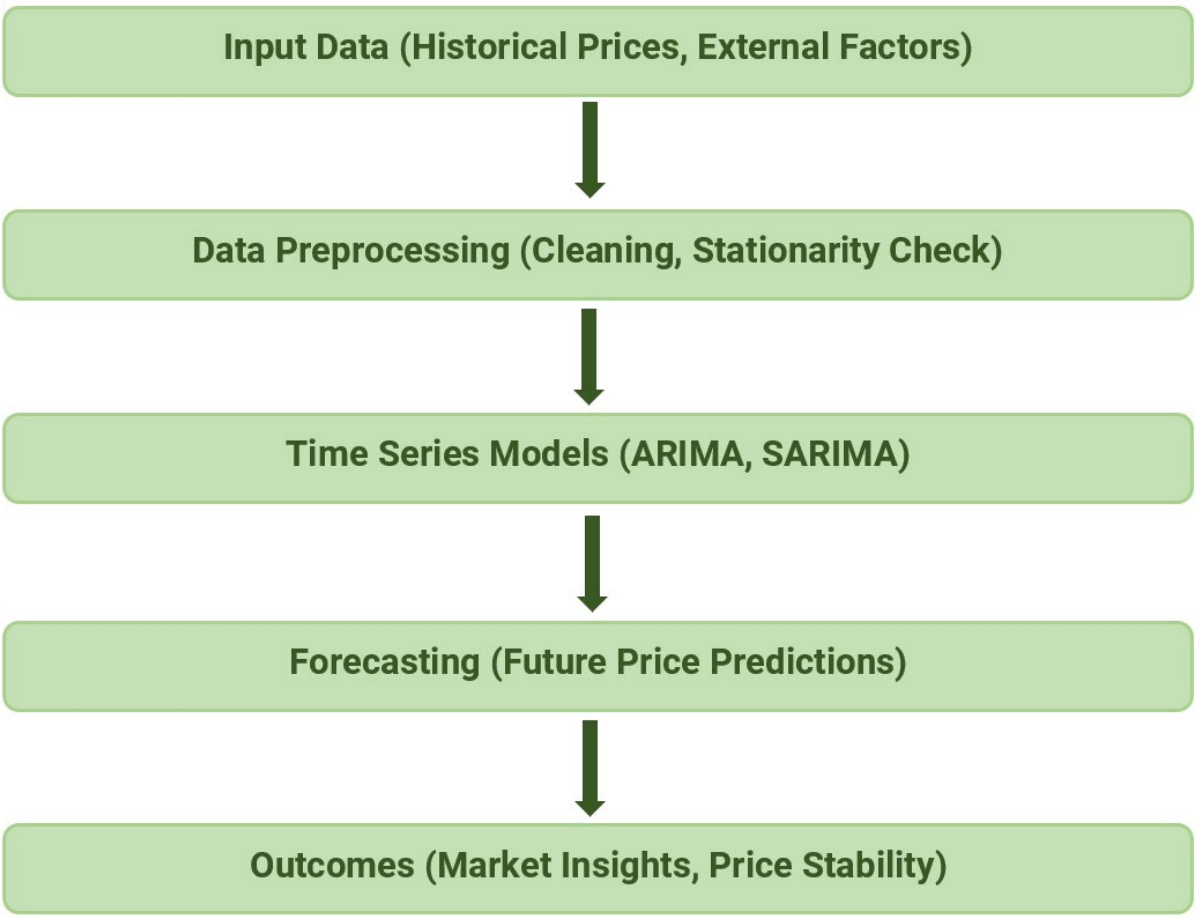

Figure 1: General Conceptual Model

4.3.3 Research Design

1)Auto Regressive Model (AR)

Autoregressive models predict future values from past values, exemplified in stock price prediction based on past performance, capturing current and previous time series relationships.

The AR model of order p:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + e_t$$

where:

- Y_t: the value of the time series at time t,
- $\phi_1, \phi_2, \ldots, \phi_p$: the finite set of weight parameters that should be estimated,
- e_t: the error term.

If Y_t is stationary and $\phi_1, \phi_2, \ldots, \phi_p$ ($\phi_p \neq 0$) are constant, then $e_t \sim N(0, \sigma_e^2)$.

2)Moving Average Models (MA)

Moving average models use past error terms for linear modeling of univariate time series. Versatile in technical analysis, they identify trends, support/resistance levels, and buying/selling opportunities based on current and past values.

The MA model of order q:

$$Y_t = e_t - \vartheta_1 e_{t-1} - \vartheta_2 e_{t-2} - \cdots - \vartheta_q e_{t-q}$$

where:

- $\vartheta_1, \vartheta_2, \ldots, \vartheta_q$: the finite set of weight parameters that should be estimated,
- $e_t \sim N(0, \sigma_e^2)$.

3)Autoregressive Moving Average Model (ARMA)

The ARMA model combines AR and MA for effective linear modeling of stationary time series with minimal parameters for comprehension and forecasting.

- AR model: regressing the variable on its own lagged values.
- MA model: modeling the error term as a linear combination of error terms occurring contemporaneously and at various times in the past.

This model is called the ARMA(p, q), where p is the order of the autoregressive part and q is the order of the moving average part. The ARMA model can be represented as:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + e_t - \vartheta_1 e_{t-1} - \vartheta_2 e_{t-2} - \cdots - \vartheta_q e_{t-q}$$

where ϕ, ϑ represent the parameters of AR and MA models respectively. $e_t \sim \text{IND}(0, \sigma_e^2)$ is distributed as an identically independent normal distribution.

5 Result and Discussion

5.1 Time Series Plot

In this section it shows the time series plot of monthly average retail price of samba rice which denotes our proposed data set.

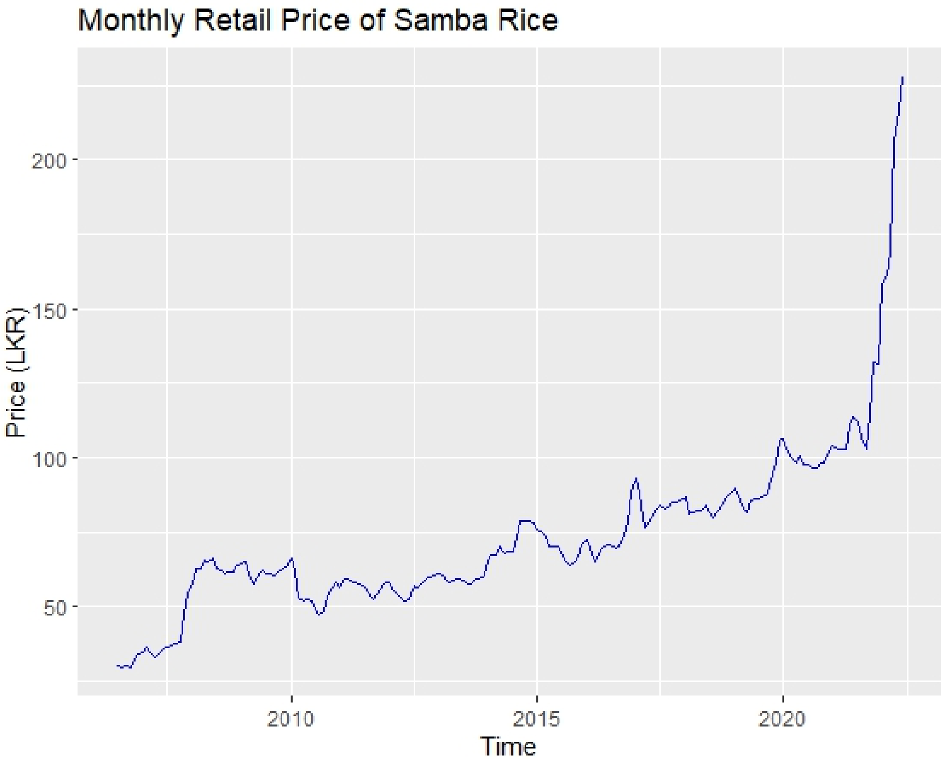

Figure 2: Retail price of samba in Colombo (2006 - 2022)

According to the above figure 2 this shows how price of Sri Lankan samba rice varies from 2006-07-15 to 2022-06-15 in Colombo city.There is a significant spike observed after 2020, attributed to the impact of COVID-19. This represents an irregular pattern in the data. To proceed with the analysis, the original dataset is divided into two parts: 80% for training and 20% for testing. The 80% training dataset is used for analysis and forecasting future values, which can then be compared against the actual values for evaluationTo show the features of the particular time series in terms of the stability of the time series in the average and variance, we draw the data using the fluctuation chart. Figure 2 shows the instability of the series in the average and variance. And we plot the ACF and PACF graphs to check weather which model is suitable for our data.Figure 2 shows that the time series appears to be non-stationary. This is because the data has a clear upward trend over time, which suggests that the mean of the series is not constant over time. So, from that we figure we can say there's an upward trend and no seasonality.

Figure 3 is a visualization used in time series analysis to break down a time series into its key components.

- Trend: The overall long-term pattern.
- Seasonality: Regular, repeating patterns (e.g., daily, monthly).
- Residual (or Noise): The random variation or unexplained component.

This plot helps identify the structure and behavior of the data, making it easier to understand patterns and make forecasts.

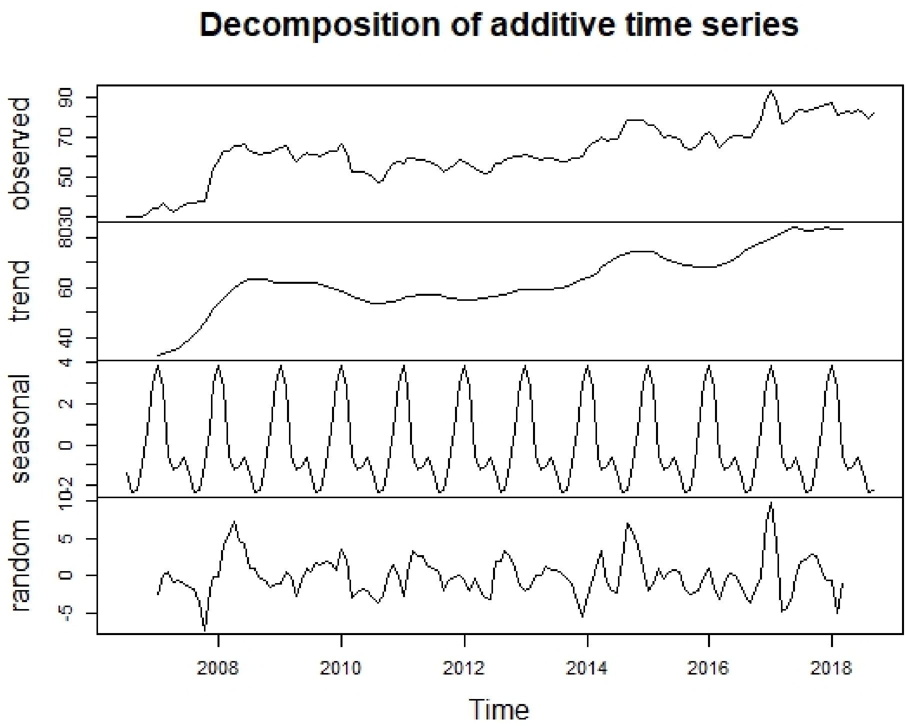

Figure 3: Decomposition plot

ACF (Autocorrelation Function) and PACF (Partial Autocorrelation Function) are statistical tools used in time series analysis to understand and describe the autocorrelation structure of a time series data set. Autocorrelation refers to the correlation of a time series with its own past and future values.

- ACF measures the overall correlation at different lags.
- PACF measures the correlation at a specific lag, excluding the effects of shorter lags.

Both ACF and PACF are often used in conjunction with each other to analyze and model time series data, especially in the context of forecasting and building predictive models. They are crucial tools in identifying the order of autoregressive (AR) and moving average (MA) processes in time series analysis. The patterns observed in ACF and PACF plots can provide insights into the structure of the time series data and guide the selection of appropriate models for forecasting.

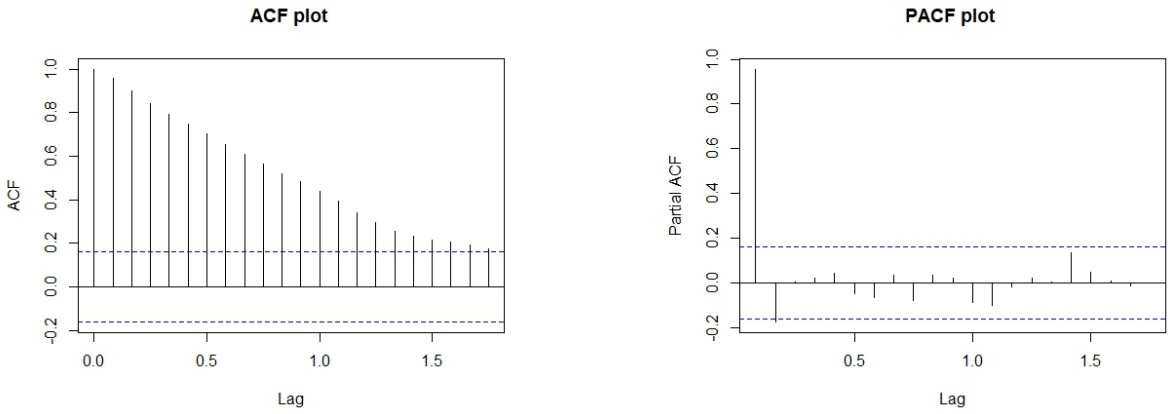

Figure 4: ACF and PACF plots for the original data

A stationary time series typically has an ACF that decreases quickly and a PACF that cuts off after a few lags. If the ACF shows a slow decay or significant correlations persist over many lags, it suggests non-stationarity in the data.

Test	P-value	Conclusion
ADF test	0.1708	Not stationary

Table 2: ADF test result

Based on the results in Table 2, it is evident that our original dataset does not exhibit stationarity according to the outcomes of ADF test. To make our data stable, we can use a method called differencing.We can ensure that by applying ADF test.

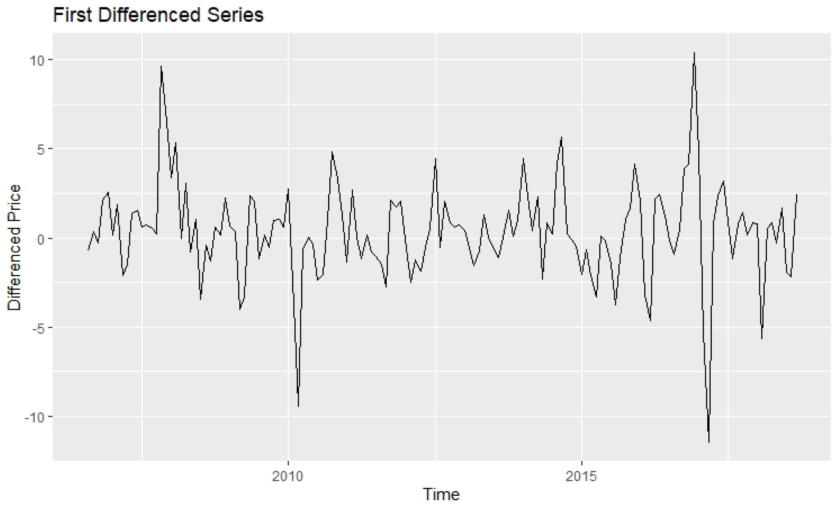

Figure 5: After 1st differencing time series plot

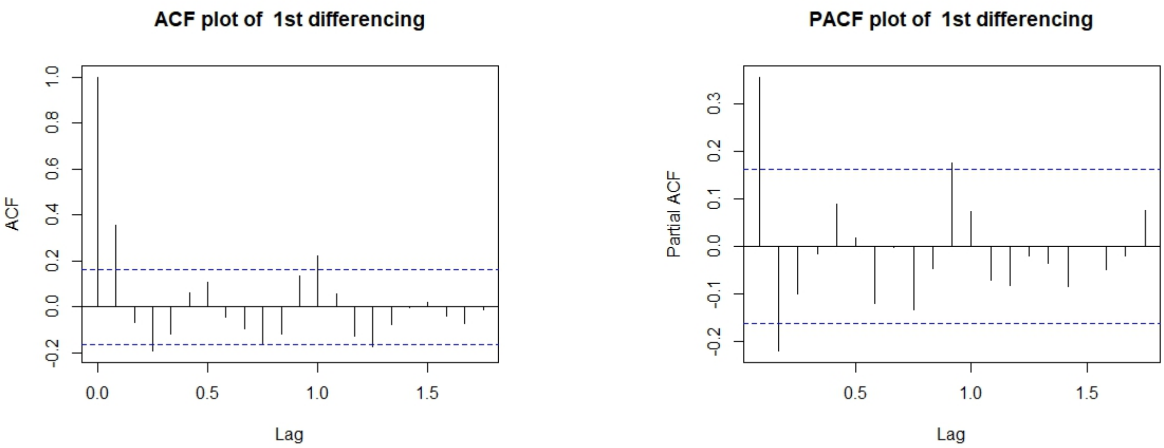

Figure 6: ACF and PACF plots after 1st differencing

Test	P-value	Conclusion
ADF test	0.01	Stationary

Table 3: ADF test result after 1st differencing

Based on the results in Table 3, it is evident that our differencing dataset is stationary according to the outcomes of ADF test.

5.2 Manual ARIMA model

The ACF plot shows that the significant correlation cuts off after the first lag (starting from lag 0), indicating Q=1. The PACF plot shows that the significant correlation cuts off after the second lag, indicating P=2. By applying first-order differencing, we achieve stationarity, so I=1. Therefore, the manual ARIMA model order is (p,d,q)=(2,1,1).

```
> print(summary(manual_arima_model))
Series: train_ts
ARIMA(2,1,1)

Coefficients:
         ar1      ar2      ma1
      0.6946  -0.3093  -0.2598
s.e.  0.2576   0.1097   0.2649

sigma^2 = 6.316:  log likelihood = -340.32
AIC=688.63   AICc=688.92   BIC=700.57

Training set error measures:
                    ME     RMSE      MAE       MPE     MAPE
Training set 0.3002626 2.478654 1.774556 0.5043844 2.892712
                  MASE        ACF1
Training set 0.2150089 -0.01695455
```

Figure 7: Manual ARIMA summary

Based on the output Figure 7 provided, the fitted ARIMA(2, 1, 1) model can be expressed as:

$$Y'_t = 0.6946Y'_{t-1} - 0.3093Y'_{t-2} + e_t + 0.2598e_{t-1}$$

where:

- Y'_t: The first-differenced series ($Y'_t = Y_t - Y_{t-1}$) to achieve stationarity.
- Y'_{t-1}: The first lag of the differenced series.
- Y'_{t-2}: The second lag of the differenced series.
- e_t: The current error term.
- e_{t-1}: The error term from the previous time step.

1. **Autoregressive (AR) Terms**:
   - $\phi_1 = 0.6946$: A strong positive influence of the previous lag (Y'_{t-1}).
   - $\phi_2 = -0.3093$: A weaker negative influence of the second lag (Y'_{t-2}).
2. **Moving Average (MA) Term**:
   - $\theta_1 = -0.2598$: A small negative contribution from the previous error (e_{t-1}).

3. **Differencing** ($d = 1$): First-order differencing was applied to make the series stationary.

This equation explains how the differenced series Y_t' is modeled based on its past values and past error terms.

5.2.1 Assumptions

1. Independence of Residuals

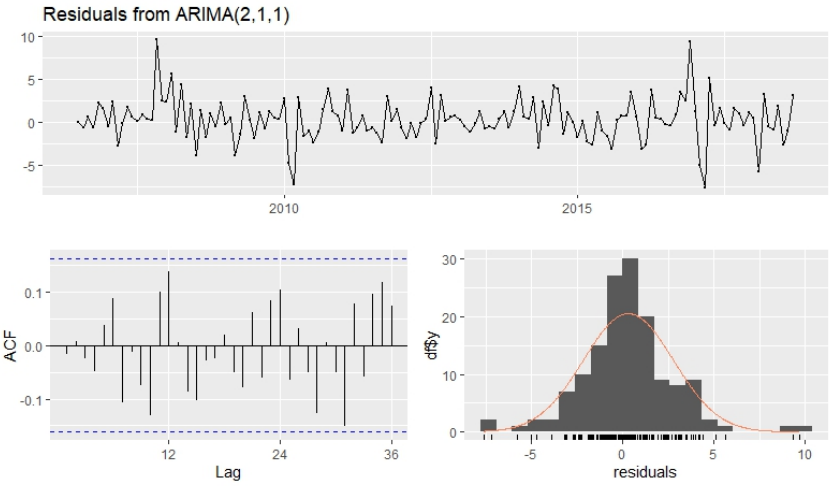

Figure 8: Manual ARIMA residuals plots

The residuals from the ARIMA model should be independent and identically distributed (i.i.d.). There should be no remaining autocorrelation in the residuals.Figure 7 look like white noise randomly scattered with no clear pattern.ACF plot show all the residuals are in significance level.

```
        Ljung-Box test

data:  Residuals from ARIMA(2,1,1)
Q* = 21.393, df = 21, p-value = 0.4352

Model df: 3.   Total lags used: 24
```

Figure 9: Ljung-Box test results

p-value is greater than 0.05, we fail to reject the null hypothesis that the residuals are not autocorrelated. This indicates that there is no significant autocorrelation in the residuals, and the model seems to be a good fit for the data.

2. Normality of Residuals

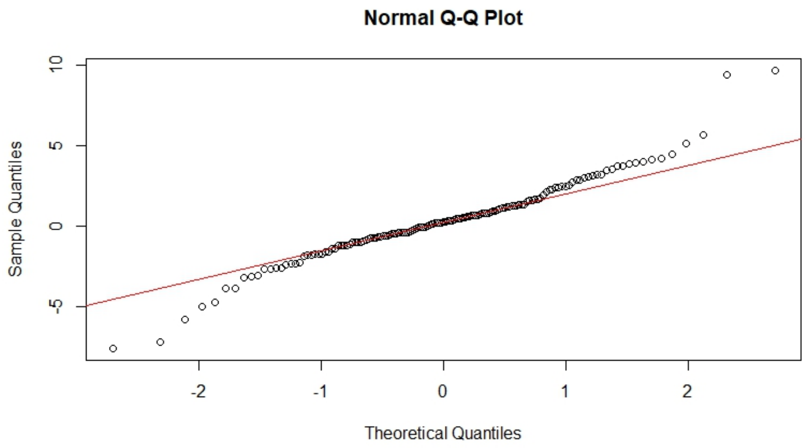

Figure 10: Normal Q-Q plot

The residuals should ideally follow a normal distribution, especially for making valid statistical inferences.The above Figure 9 shows approximately normaly distributed.

5.2.2 Forecasting

In this section, we aim to forecast the monthly average retail price of Samba rice in Colombo for the next year starting from September 2018 (12 months) using the ARIMA(2,1,1) model.

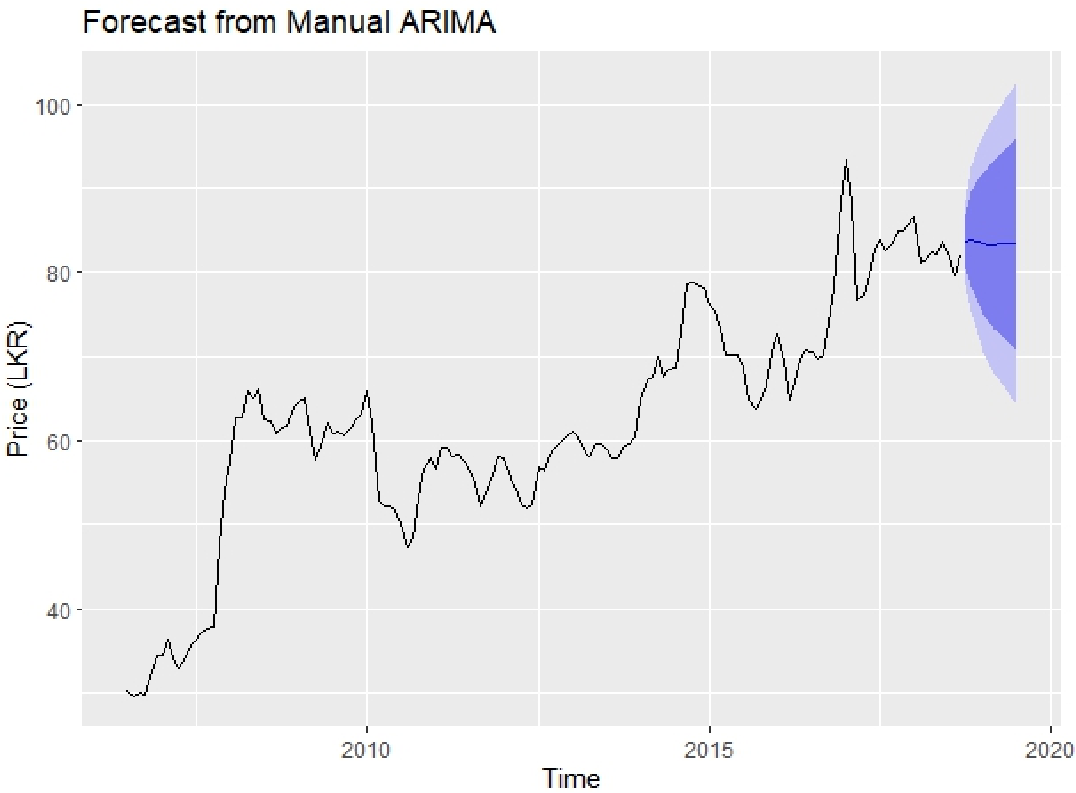

Figure 11: Forecast plot

Month	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Oct 2018	83.63085	80.41021	86.85150	78.70530	88.55640
Nov 2018	83.96657	78.33416	89.59899	75.35253	92.58061
Dec 2018	83.71705	76.44663	90.98746	72.59791	94.83619
Jan 2019	83.43991	75.06948	91.81034	70.63844	96.24138
Feb 2019	83.32458	74.11407	92.53510	69.23832	97.41085
Mar 2019	83.33019	73.37786	93.28252	68.10941	98.55097
Apr 2019	83.36975	72.71161	94.02789	67.06953	99.66997
May 2019	83.39549	72.05754	94.73345	66.05558	100.73540
Jun 2019	83.40114	71.41438	95.38790	65.06897	101.73331
Jul 2019	83.39710	70.79496	95.99924	64.12379	102.67040
Aug 2019	83.39255	70.20588	96.57921	63.22528	103.55981
Sep 2019	83.39063	69.64569	97.13558	62.36956	104.41171

Table 4: Forecast Table with 80% and 95% Confidence Intervals

5.3 Auto ARIMA model

The auto.arima function in R automatically identifies the best-fitting ARIMA model for a given time series. It uses a stepwise search algorithm to select the optimal combination of parameters (p, d, q) by minimizing the Akaike Information Criterion (AIC), AIC corrected (AICc), or Bayesian Information Criterion (BIC).

```
> print(summary(auto_arima_model))
Series: train_ts
ARIMA(0,1,1)

Coefficients:
         ma1
      0.4148
s.e.  0.0700

sigma^2 = 6.385:  log likelihood = -342.1
AIC=688.19   AICc=688.28   BIC=694.16

Training set error measures:
                    ME     RMSE      MAE      MPE     MAPE
Training set 0.2552476 2.509628 1.794561 0.437988 2.920579
                  MASE       ACF1
Training set 0.2174327 0.01299262
```

Figure 12: Auto ARIMA summary

The fitted ARIMA(0,1,1) model can be expressed as:

$$Y'_t = e_t - 0.4148 e_{t-1}$$

where:

- $Y'_t = Y_t - Y_{t-1}$: The first-differenced series, representing the changes in the original time series.
- e_t: The current error term (random noise).
- e_{t-1}: The error term from the previous time step, weighted by $\theta_1 = 0.4148$.

This equation indicates that the differenced series Y'_t depends on the current and previous error terms.

5.3.1 Assumptions

1. Independent residuals

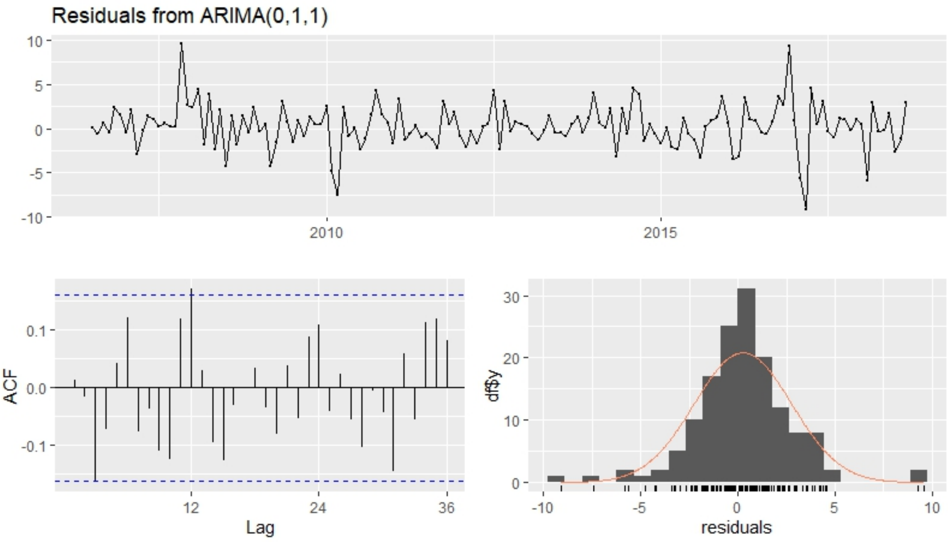

Figure 13: Auto ARIMA residuals plots

The residuals from the ARIMA model should be independent and identically distributed (i.i.d.). There should be no remaining autocorrelation in the residuals.Figure 13 look like white noise randomly scattered with no clear pattern.ACF plot show all the residuals are in significance level.

```
        Ljung-Box test

data:  Residuals from ARIMA(0,1,1)
Q* = 30.313, df = 23, p-value = 0.1406

Model df: 1.   Total lags used: 24
```

Figure 14: Ljung-Box results Auto ARIMA

p-value is greater than 0.05, we fail to reject the null hypothesis that the residuals are not autocorrelated. This indicates that there is no significant autocorrelation in the residuals, and the model seems to be a good fit for the data.

2. Normality of Residuals

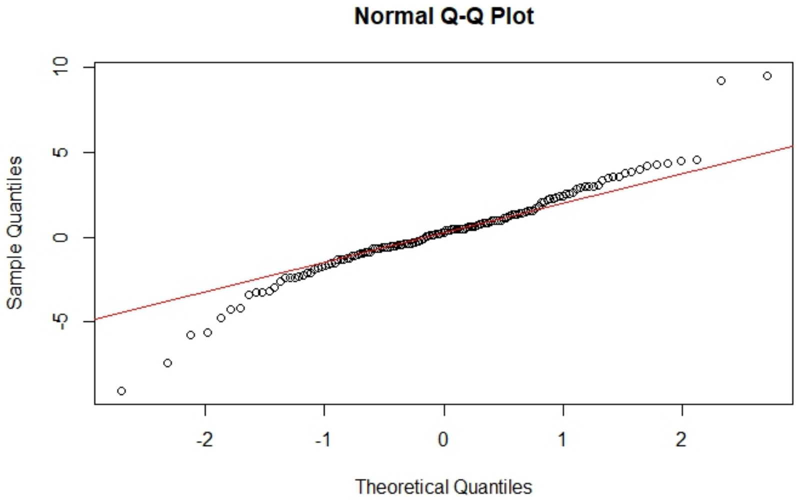

Figure 15: Normal Q-Q plot Auto ARIMA residuals

The residuals should ideally follow a normal distribution, especially for making valid statistical inferences.The above Figure 15 shows approximately normaly distributed.

5.3.2 Forecasting

In this part, we aim to forecast the monthly average retail price of Samba rice in Colombo for the next year starting from September 2018 (12 months) using the ARIMA(0,1,1) model.

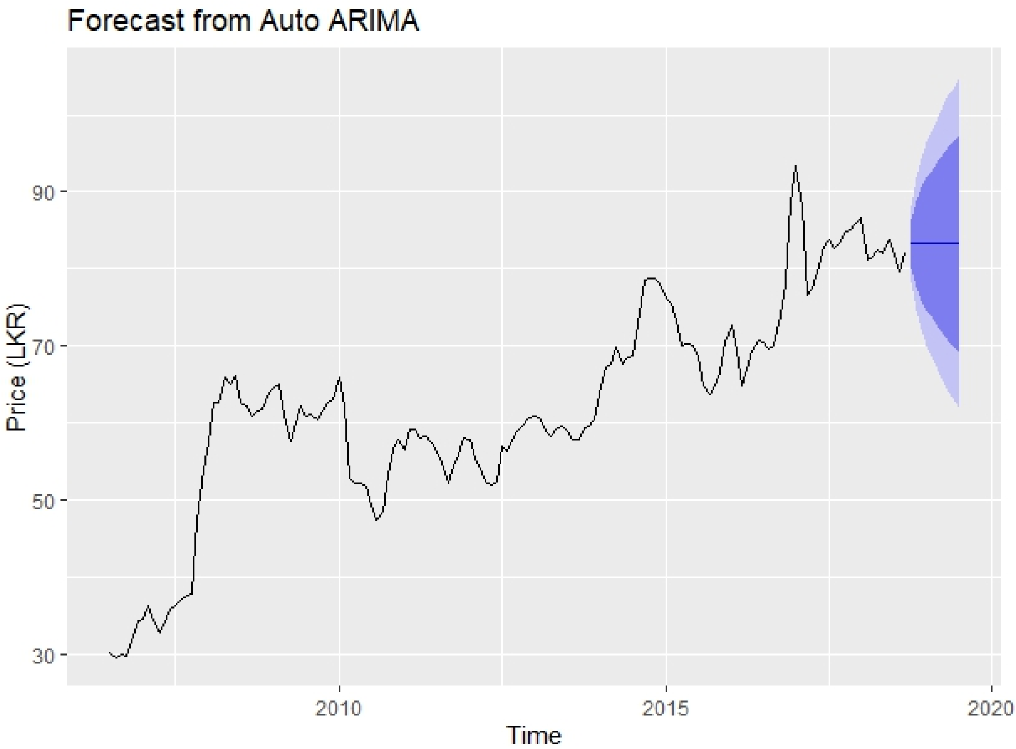

Figure 16: Forecast plot

Month	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Oct 2018	83.26378	80.02546	86.50211	78.31120	88.21637
Nov 2018	83.26378	77.65328	88.87428	74.68326	91.84431
Dec 2018	83.26378	76.02025	90.50731	72.18576	94.34181
Jan 2019	83.26378	74.69292	91.83465	70.15578	96.37179
Feb 2019	83.26378	73.54521	92.98236	68.40050	98.12706
Mar 2019	83.26378	72.51940	94.00816	66.83167	99.69590
Apr 2019	83.26378	71.58334	94.94423	65.40009	101.12748
May 2019	83.26378	70.71692	95.81064	64.07501	102.45255
Jun 2019	83.26378	69.90659	96.62098	62.83571	103.69185
Jul 2019	83.26378	69.14267	97.38489	61.66741	104.86016
Aug 2019	83.26378	68.41802	98.10955	60.55915	105.96842
Sep 2019	83.26378	67.72713	98.80044	59.50252	107.02505

Table 5: Forecast Table with 80% and 95% Confidence Intervals

5.4 Comparison of Original Data and Auto ARIMA's Forecast

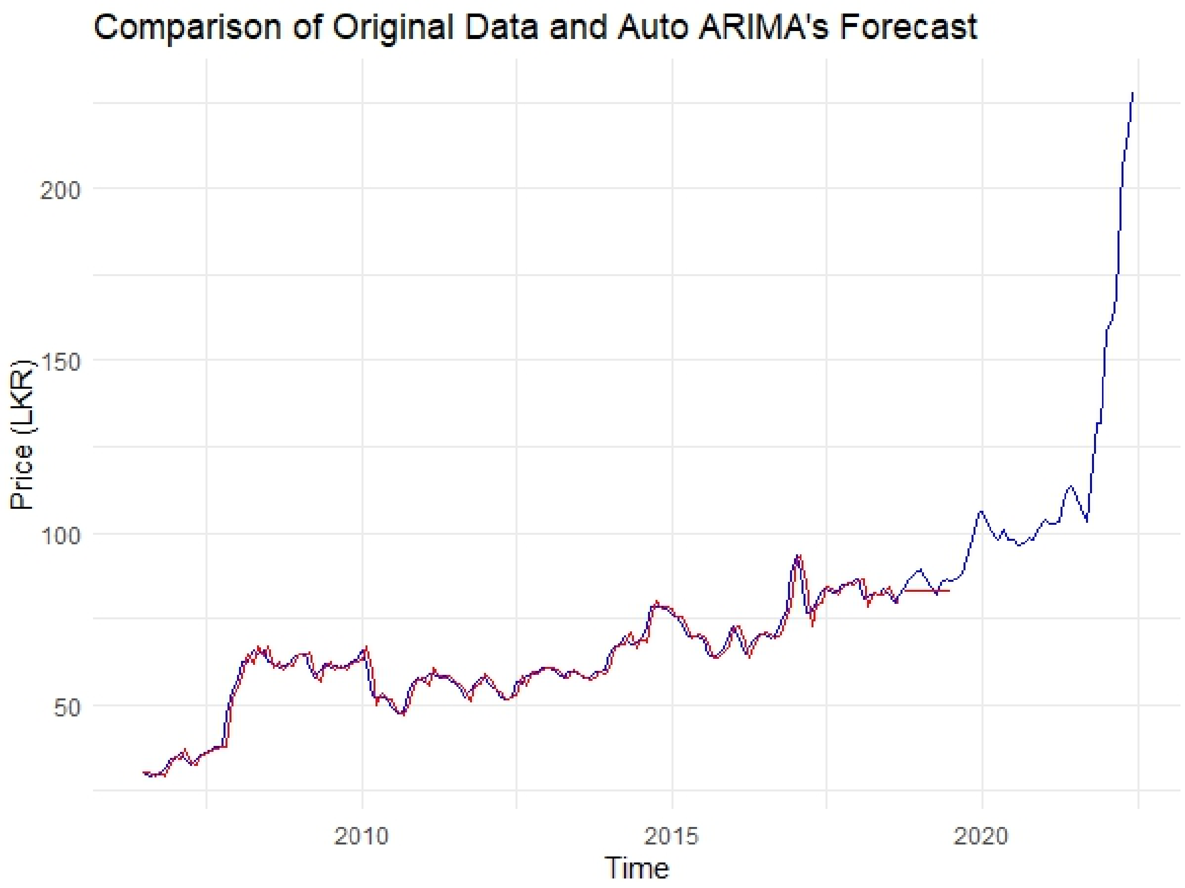

Figure 17: Forecast plot vs original data

Comparison of the original time series data with the forecast generated by the Auto ARIMA model.the graph compares the original data (blue line) with the forecast generated by the Auto ARIMA model (red line). The model closely tracks historical trends and provides future projections based on observed patterns.

5.5 Comparison of Original Data and Manual ARIMA's Forecast

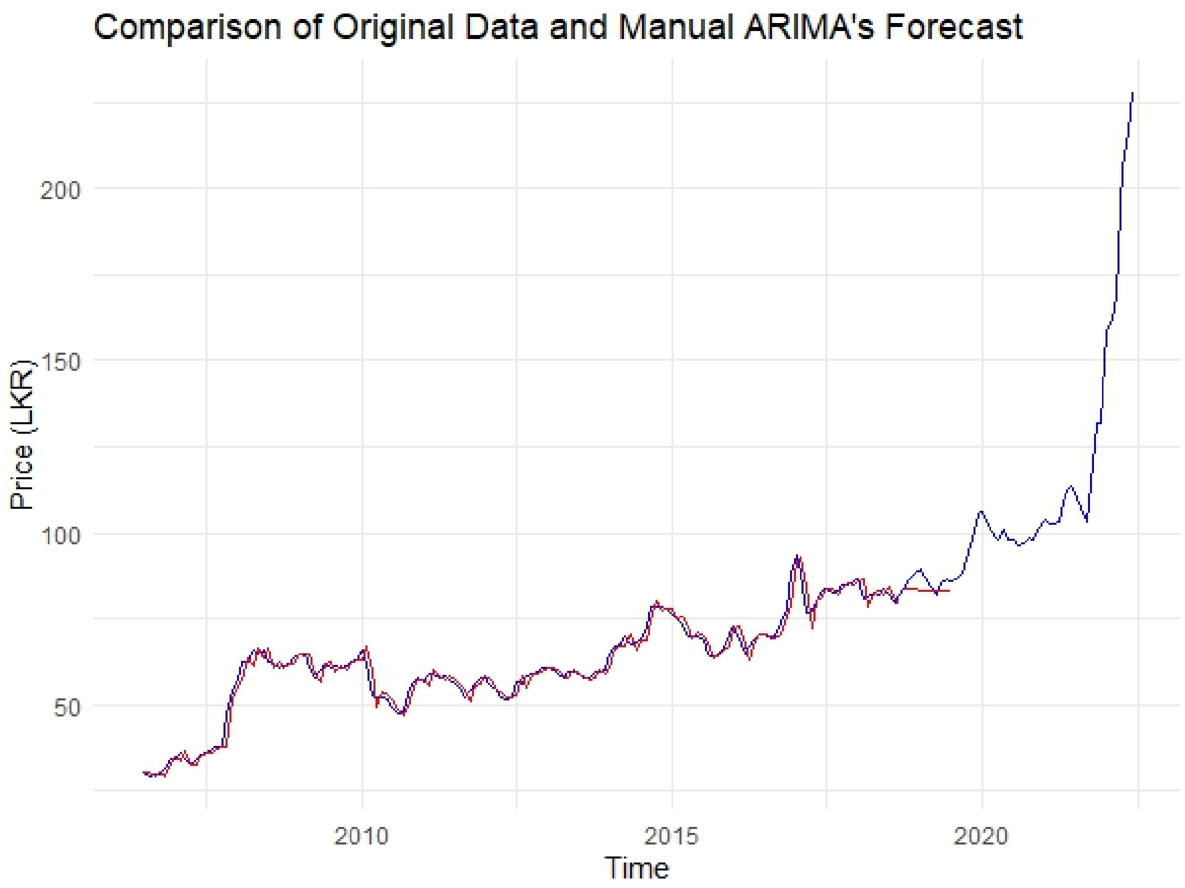

Figure 18: Forecast plot vs original data

This graph illustrates the comparison between the original time series data (blue line) and the forecast produced by the Manual ARIMA model (red line). The model effectively captures historical trends and projects future values based on these patterns.

5.6 Auto ARIMA and Manual ARIMA models comparison

Table 6 summarizes the outcomes of the different criteria for both the Auto ARIMA and Manual ARIMA models.

Criterion	**Auto ARIMA**	**Manual ARIMA**	**Conclusion**
Model Simplicity	Simpler (ARIMA(0,1,1))	More Complex (ARIMA(2,1,1))	**Auto ARIMA** is simpler.
AIC	688.19	688.63	**Auto ARIMA** is more efficient.
BIC	694.16	700.57	**Auto ARIMA** is more efficient.
RMSE	3.58	3.39	**Manual ARIMA** has slightly better accuracy.
MAPE	3.66	3.43	**Manual ARIMA** has slightly better accuracy.
Residual Analysis (ACF1)	0.01299	-0.01695	Both models have nearly white noise residuals.

Table 6: Comparison of Auto ARIMA and Manual ARIMA Models based on Different Criteria.

Final Recommendation

Based on the comparison of both models using the different criteria, we can conclude the following:

- **Auto ARIMA** is the more efficient model due to its lower AIC and BIC, and it is simpler in terms of model complexity.
- **Manual ARIMA** provides slightly better prediction accuracy (lower RMSE and MAPE), but it is more complex with additional parameters.

Decision:

- If **model simplicity** and **efficiency** are more important, **Auto ARIMA** should be the preferred choice.
- If **prediction accuracy** is the top priority, and you are willing to accept a slightly more complex model, then **Manual ARIMA** is the better option.

6 Conclusion

This study analyzed the monthly average retail prices of Samba rice in Colombo, Sri Lanka, using time series modeling techniques to provide accurate forecasts and support decision-making for market stabilization. The findings demonstrate the robustness of the manually developed ARIMA (2,1,1) model, which was identified as the most effective for forecasting compared to the Auto ARIMA (0,1,1) model. Key insights from the analysis include:

1. **Model Selection and Performance:**
   The manual ARIMA (2,1,1) model proved superior for forecasting as it captured the underlying data patterns more effectively. This model met all assumptions, including no autocorrelation in residuals and normally distributed errors, ensuring its reliability and robustness.

2. **Forecasting Accuracy:**
   The manual ARIMA model provided reliable 12-month forecasts with confidence intervals, offering critical insights for market trends. These forecasts are valuable for policymakers, traders, and farmers to make informed decisions regarding market regulation and planning.

3. **Economic Implications:**
   The study revealed significant price volatility, particularly after 2020 due to external shocks like the COVID-19 pandemic. This highlights the importance of using manual ARIMA models for more precise forecasting in dynamic and uncertain market conditions.

Implications:
The findings underscore the efficacy of manually developed ARIMA models in forecasting agricultural commodity prices. This approach provides stakeholders with reliable predictions that enable better risk management, price stabilization, and informed policy-making. By utilizing the manual ARIMA model, Sri Lanka's rice market can achieve greater stability, benefiting both consumers and producers.

Future research could consider incorporating external variables such as economic policies, climatic factors, and global market trends to further refine the forecasting models and enhance their predictive capabilities.

References

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Appendix

R Code for Time Series Forecasting of Samba Rice Prices

```
# Load libraries
library(fpp2)
library(tseries)
library(forecast)
library(ggplot2)
library(FinTS)
library(gridExtra)

# 1. Load and Inspect the Data
rice_data <- read.csv("Cleaned_Rice_Data.csv", header =
   TRUE) # Replace with your file path
rice_data$date <- as.Date(rice_data$date, format = "%m/%d/%
   Y")
rice_data$price <- as.numeric(rice_data$price)

# Check for missing or invalid data
rice_data <- rice_data[!is.na(rice_data$date) & !is.na(
   rice_data$price), ]
rice_data$price <- as.numeric(rice_data$price)
missing_data <- sum(is.na(rice_data$price) & is.na(
   rice_data$date))
if (missing_data == 0) print("No Missing Values") else
   print("There are Missing Values")

# Check data summary and structure
print(summary(rice_data))
print(str(rice_data))

# Ensure there are valid observations
if (nrow(rice_data) == 0) {
  stop("No valid observations in the dataset.")
}

# Create a time series object (assuming monthly data
   starting July 2006)
ts_data <- ts(rice_data$price, start = c(2006, 7),
```

```
  frequency = 12)

# Plot the original time series
autoplot(ts_data, main = "Original Time Series plot '2006-07-15 - 2022-06-15'", ylab = "Price (LKR)")

# 2. Split Data into Training and Test Sets
n <- length(ts_data)
train_size <- floor(0.8 * n)
train_ts <- window(ts_data, end = c(2006 + (train_size - 1) %/% 12, (train_size - 1) %% 12 + 1))
test_ts <- window(ts_data, start = c(2006 + train_size %/% 12, train_size %% 12 + 1))

# 3. Decompose and Analyze Seasonality
acf(train_ts, main="ACF plot")
pacf(train_ts, main="PACF plot")
decomposed_ts <- decompose(train_ts)
plot(decomposed_ts)

# Check for seasonality using a statistical test
seasonality_test <- isSeasonal(train_ts)
print(paste("Is the series seasonal? ", seasonality_test))

# 4. Stationarity Check
adf_test <- adf.test(train_ts)
print(adf_test)
if (adf_test$p.value <= 0.05) {
  print("The series is stationary.")
} else {
  print("The series is not stationary. Differencing is required.")
}

# 5. Differencing to Ensure Stationarity
ts_diff1 <- diff(train_ts)
autoplot(ts_diff1, main = "First Differenced Series", ylab = "Differenced Price")

```

```
adf_test_diff1 <- adf.test(ts_diff1)
print(adf_test_diff1)
if (adf_test_diff1$p.value <= 0.05) {
  print("The differenced series is stationary.")
} else {
  print("Further differencing might be required.")
}

# 6. ACF and PACF Analysis
acf_plot <- acf(ts_diff1, main='ACF plot of 1st differencing')
pacf_plot <- pacf(ts_diff1, main='PACF plot of 1st differencing')

# 7. Fit ARIMA Models
# Automatic ARIMA
auto_arima_model <- auto.arima(train_ts, seasonal = F, trace = TRUE)

# Manual ARIMA
manual_arima_model <- Arima(train_ts, order = c(2, 1, 1))

# Print model summaries
print(summary(auto_arima_model))
print(summary(manual_arima_model))

# 8. Assumption Checks
# Residual analysis for Auto ARIMA
checkresiduals(auto_arima_model)

shapiro_test_auto <- shapiro.test(residuals(auto_arima_model))
print(shapiro_test_auto)
plot(residuals(auto_arima_model), main = "Residuals of Auto ARIMA Model")
qqnorm(residuals(auto_arima_model))
qqline(residuals(auto_arima_model), col = 'red')

# Ljung-Box test for autocorrelation of residuals
```

```
ljung_box_auto <- Box.test(residuals(auto_arima_model), lag = 20, type = 'Ljung-Box')
print(ljung_box_auto)

acf(residuals(auto_arima_model), main = "ACF of Residuals")
pacf(residuals(auto_arima_model), main = "PACF of Residuals")

# Residual analysis for Manual ARIMA
checkresiduals(manual_arima_model)
plot(residuals(manual_arima_model), main = "Residuals of Manual ARIMA Model")
qqnorm(residuals(manual_arima_model))
qqline(residuals(manual_arima_model), col = 'red')
acf(residuals(manual_arima_model), main = "ACF of Residuals")
pacf(residuals(manual_arima_model), main = "PACF of Residuals")

# 9. Model Forecasting
forecast_auto <- forecast(auto_arima_model, h = 12)
forecast_manual <- forecast(manual_arima_model, h = 12)

autoplot(forecast_auto) +
  ggtitle("Forecast from Auto ARIMA") +
  xlab("Time") +
  ylab("Price (LKR)")

autoplot(forecast_manual) +
  ggtitle("Forecast from Manual ARIMA") +
  xlab("Time") +
  ylab("Price (LKR)")

# 10. Model Evaluation
rmse_auto <- sqrt(mean((forecast_auto$mean - test_ts)^2))
mape_auto <- mean(abs((forecast_auto$mean - test_ts) / test_ts)) * 100
print(paste("Auto ARIMA - RMSE:", round(rmse_auto, 2), " MAPE:", round(mape_auto, 2)))
```

```
rmse_manual <- sqrt(mean((forecast_manual$mean - test_ts)^2))
mape_manual <- mean(abs((forecast_manual$mean - test_ts) / test_ts)) * 100
print(paste("Manual ARIMA - RMSE:", round(rmse_manual, 2), "MAPE:", round(mape_manual, 2)))

# 11. Combine forecast with original data for comparison
autoplot(ts_data, series = "Original Data", color = "blue") +
  autolayer(fitted(manual_arima_model), series = "Fitted (Manual ARIMA)", color = "red") +
  autolayer(forecast(manual_arima_model, h = 10)$mean, series = "Forecast (Manual ARIMA)", color = "red") +
  ggtitle("Comparison of Original Data and Manual ARIMA's Forecast") +
  xlab("Time") +
  ylab("Price (LKR)") +
  scale_color_manual(values = c("blue", "red", "red")) +
  theme_minimal() +
  theme(legend.title = element_blank())

autoplot(ts_data, series = "Original Data", color = "blue") +
  autolayer(fitted(auto_arima_model), series = "Fitted (Auto ARIMA)", color = "red") +
  autolayer(forecast(auto_arima_model, h = 10)$mean, series = "Forecast (Auto ARIMA)", color = "red") +
  ggtitle("Comparison of Original Data and Auto ARIMA's Forecast") +
  xlab("Time") +
  ylab("Price (LKR)") +
  scale_color_manual(values = c("blue", "red", "red")) +
  theme_minimal() +
  theme(legend.title = element_blank())
```

Listing 1: R Code for Time Series Forecasting of Samba Rice Prices